



Solving linear equations where brackets are used

Task A: Using reciprocals to solve equations with brackets

1) Solve these equations in two different ways. Firstly, by expanding the bracket as a first step. Secondly, by multiplying by the reciprocal.

a) $2(x + 7) = 30$
 $2x + 14 = 30$
 $2x = 16$
 $x = 8$

$2(x + 7) = 30$
 $\frac{2(x + 7)}{2} = \frac{30}{2}$
 $x + 7 = 15$
 $x = 8$

b) $21 = 7(2x + 7)$
 $21 = 14x + 49$
 $-28 = 14x$
 $x = -2$

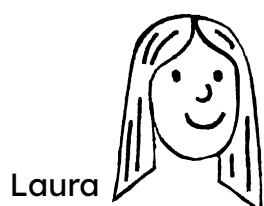
$21 = 7(2x + 7)$
 $\frac{21}{7} = \frac{7(2x + 7)}{7}$
 $3 = 2x + 7$
 $x = -2$

2) Solve by expanding first:

a) $5(x - 1) = 3(3x - 11)$
 $5x - 5 = 9x - 33$
 $-5 = 4x - 33$
 $28 = 4x$
 $x = 7$

b) $3(x - 1) = 5(3x - 11)$
 $3x - 3 = 15x - 55$
 $3x + 52 = 15x$
 $52 = 12x$
 $x = \frac{52}{12} = \frac{13}{3}$

3) Laura and Izzy have solved an equation by removing the brackets.



$5(x - 10) = 45$
 $5x - 10 = 45$
 $5x = 55$
 $x = 11$



$5(x - 10) = 45$
 $x - 10 = 9$
 $x = 19$



Izzy

Only one pupil is correct.

Write a sentence explaining why the right method worked and write a sentence explaining what went wrong in the other one.

Izzy is correct. She has multiplied both sides by the reciprocal $\frac{1}{5}$

$\frac{5(x - 10)}{5} = \frac{45}{5}$ leaves the simple equation $x - 10 = 9$

Laura is incorrect. She has simply erased the brackets rather than multiplied them out; you have to make sure both terms inside the bracket are multiplied. $5(x - 10)$ should multiply out to $5x - 50$



Task B: Efficient methods to solve equations

1) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$\begin{aligned} 5(x + 13) &= 40 \\ 5x + 65 &= 40 \\ 5x &= -25 \\ x &= -5 \end{aligned}$$

Multiplying by the reciprocal first:

You may have written, "For this equation $5(x + 13) = 40$ multiplying by the reciprocal was the most efficient method. In one step it gave us a simple equation to solve."

$$\begin{aligned} x + 13 &= 8 \\ x &= -5 \end{aligned}$$

2) Solve both of these equations twice using two different methods.

a) $2(x + 6) = 10(x - 2)$

$$\begin{aligned} 2x + 12 &= 10x - 20 \\ 12 &= 8x - 20 \\ 32 &= 8x \\ x &= 4 \end{aligned}$$

$$\begin{aligned} 20(x + 2) &= 5(7x - 1) \\ 20x + 40 &= 35x - 5 \\ 40 &= 15x - 5 \\ 45 &= 15x \\ x &= 3 \end{aligned}$$

$$\begin{aligned} 2(x + 6) &= 10(x - 2) \\ \frac{2(x + 6)}{2} &= \frac{10(x - 2)}{2} \\ x + 6 &= 5(x - 2) \\ x + 6 &= 5x - 10 \\ 16 &= 4x \\ x &= 4 \end{aligned}$$

$$\begin{aligned} 20(x + 2) &= 5(7x - 1) \\ \frac{20(x + 2)}{5} &= \frac{5(7x - 1)}{5} \\ 4(x + 2) &= 7x - 1 \\ 4x + 8 &= 7x - 1 \\ 9 &= 3x \\ x &= 3 \end{aligned}$$

3) Solve:

a) $\frac{3}{5}(7x - 4) = 21$

$$\begin{aligned} \frac{5}{3} \times \frac{3}{5}(7x - 4) &= \frac{5}{3} \times 21 \\ 7x - 4 &= 35 \\ 7x &= 39 \\ x &= \frac{39}{7} \end{aligned}$$

b) $\frac{1}{5}(5x - 15) = \frac{3}{2}(x + 4)$

$$\begin{aligned} x - 3 &= \frac{3}{2}(x + 4) \\ \frac{2}{3} \times (x - 3) &= \frac{2}{3} \times \frac{3}{2}(x + 4) \\ \frac{2x}{3} - 2 &= x + 4 \\ -6 &= \frac{1}{3}x \\ x &= -18 \end{aligned}$$

or... $\frac{1}{5}(5x - 15) = \frac{3}{2}(x + 4)$

$$\begin{aligned} 10 \times \frac{1}{5}(5x - 15) &= 10 \times \frac{3}{2}(x + 4) \\ \frac{10}{5}(5x - 15) &= \frac{30}{2}(x + 4) \\ 2(5x - 15) &= 15(x + 4) \\ 10x - 30 &= 15x + 60 \\ -90 &= 5x \\ x &= -18 \end{aligned}$$

